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SUMMARY

- 2+ years of experience in object-oriented software development, supporting building, testing and deployment for pre-production apps and cloud-based applications used across 5,700+ locations in Canada.
- Developed, scaled and unit tested a pre-production application using Java, TypeScript/JavaScript and PostgreSQL, resulting in a 60% reduction in bugs.
- Tested and delivered enterprise applications hosted on Azure across 10 major releases by leveraging Continuous Integration and Continuous Delivery (CI/CD) pipelines.
- Independently designed and developed 3 projects using diverse technology stacks, including React/Node.js/Express/PostgreSQL, TypeScript/JavaScript/Java/Python/MySQL, and C++/MySQL. Containerized with Docker and deployed to a Linux-based hosting platform.

PROFESSIONAL EXPERIENCE

PrimeGO

Software Developer (Contract)

Remote/Kingston
Feb 2024 - Dec 2024

PrimeGO is an early-stage local delivery startup preparing to launch its on-demand logistics platform and aiming to raise \$10M in 2025.

- One of 5 Software Developers reporting to the Technical Lead supporting PrimeGO in developing its MVP into a production-ready product.
- Developed and scaled a Spring Boot MVC web application using Java, building secure RESTful APIs, integrating with JPA/Hibernate for database operations, and implementing role-based access control.
- Developed and integrated RESTful API calls in Angular, utilizing RxJS for asynchronous data handling, improving data fetch efficiency by 40%.
- Built reusable and responsive UI components using HTML, CSS and TypeScript/JavaScript in VSCode, ensuring clean, maintainable code and improving front-end performance by 30% through efficient data binding, component lifecycle management and modern styling practices.
- Optimized data access layers in backend services by designing efficient SQL, improving query performance and reducing database response times by 25% in PostgreSQL environments.
- Designed and automated 100+ unit and integration tests using Spring Boot Test, MockMVC, Mockito to validate business logic and API responses, resulting in a 60% reduction in critical bug reports.
- Followed Agile methodologies using JIRA for task management and leverage Git/GitHub for efficient version control and collaboration.

Innovapost

DevOps Engineer (Co-op)

Ottawa, ON
May 2023 – Aug 2023

Innovapost was a technology services provider for the public sector. With 1,000+ employees, it completed 100+ major projects for clients across Canada.

- Served as one of 7 DevOps Engineers reporting to the Solution Support Manager, providing support for a cloud-based Retail Point of Sale application used by 5700+ locations nationwide.
- Deployed new builds to 10+ testing environments using Jenkins Continuous Integration and Continuous Delivery (CI/CD) pipelines, Ansible Infrastructure-as-Code (IaC), Git and Bitbucket, ensuring 5 deployments per week and resolving build failures, resulting in nearly 400 successful deployments delivered.

- Managed Azure-hosted IaaS microservices and performed weekly batch processing of 10+ Windows virtual machines, totalling more than 160 batches.
- Partnered with development teams to diagnose and resolve 5 critical production issues, leveraging Azure Log Analytics to analyze logs from services including the POS application and API endpoints.
- Developed 5 Azure Alerts using KQL queries to proactively detect and monitor resolved production issues.
- Documented 5 resolved issues and maintained 3 deployment version tracking documents for production and testing environments on Confluence.

QA Tester (Co-op)

Jan 2023 - Apr 2023

- One of 6 QA testers reporting to the Technical Lead supporting testing of a Retail Vendor POS integrated with 15+ services and 8 devices.
- Revamped and integrated a dormant automation regression testing project for desktop application using Visual Basic, decreasing the testing time by 70%.
- Conducted automated and manual testing (regression, end-to-end, and compatibility) on desktop and mobile applications across 5 major releases, executing nearly 420 test cases.
- Identified 2 major defects through automated regression testing, preventing a potential \$10,000 in losses.
- Conducted manual post-release testing for 5 major releases and 1 hot-fix release.
- Participated in daily Agile/Scrum meetings with cross-functional teams, providing task updates and ensuring timely, high-quality software releases.

Algonquin College Peer Tutor

Ottawa, ON
Sep 2022 - Dec 2022

- As one of 5 peer tutors who achieved an A or higher in coursework, assisted 20+ peers in Object-oriented Programming (Java), Database Systems (SQL), Linux Operating System, and Web Development across the program.

PROJECTS

JobTracker

Present

JobTracker a full-stack job application tracker to organize and monitor job search progress.

- Built a RESTful API with Node.js and Express, implemented a responsive React frontend for managing applications, and integrated a MySQL database to persist user data.
- Aimed to streamline job tracking, improve productivity, and support status updates and search functionality in real-time.

ActiveTO

ActiveTO is a live full stack web application that provides browsing, searching and filtering drop-in activities at 152 City of Toronto recreation centers, as well as real-time directions to facilities.

Back-end

Mar 2024 - Jun 2024

- Designed and developed a REST API with 17 HTTP endpoints using Spring Boot MVC, built with Maven, and documented with Swagger/OpenAPI for clear API specifications.
- Developed a Python ETL pipeline to extract data in JSON from 3 sources in the Toronto Open Data API, process 40,000+ records and store the latest activity data weekly.
- Designed and implemented a MySQL schema with 12 tables to store processed data and support future feature expansion.
- Optimized SQL queries, reducing API response times by 20% through indexing strategies.

Front-end

Jul 2024 - Aug 2024

- Designed and developed a responsive front-end application using Angular, NgRx and Webpack, featuring 8 UI components and 14 dynamic rendering components.
- Integrated the ActiveTO REST API to fetch activities and the Google Maps API to display recreation center locations with real-time directions.
- Utilized MVVM pattern and integrated Material UI for responsive design and ensuring consistency across different devices and screen sizes.
- Developed state management with 34 actions and 2 reducers using NgRx, enhancing application performance and maintainability.

Deployment and Documentation

Aug 2024

- Developed a Dockerfile to containerize the REST API, ETL pipeline, and MySQL database into 3 separate containers for improved portability and scalability.
- Developed a CI/CD pipeline using GitHub Actions for seamless deployment to a Linux-based hosting platform on DigitalOcean.
- Developed a Nginx configuration script to implement a proxy service.
- Created 2 detailed documentation covering component architecture diagrams, database schema, API integration, and the CI/CD pipeline.

TravelExpensesTracker

Sep 2023

- Developed a C++ command-line application to track and manage travel expenses, following the MVC design pattern for clean separation of concerns.
- Built and compiled using CMake to ensure portability and maintainability across environments.

EDUCATION

Algonquin College
Computer Programming

Ottawa, ON
2023

Northwestern A&F University
Biotechnology

Xi'an, China
2014

CERTIFICATE

- AWS Certified Cloud Practitioner

TECHICAL SKILLS

- **Front-end Development:** TypeScript, JavaScript, Angular, HTML, CSS
- **Back-end Development:** Java, C++, Spring, Spring Boot, Python, Bash
- **Test Automation:** JUnit, Spring Boot Test, MockMVC, Mocha, Jest
- **Databases:** PostgreSQL, MySQL, MS SQL Server, SQLite
- **DevOps & CI/CD:** Azure DevOps, GitHub Actions, Jenkins, Docker
- **Version Control:** Git, GitHub, Bitbucket
- **Development Management:** JIRA, Confluence, Agile/Scrum